Circuit investigation

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use measurements to test a circuit model safely.

SUCCESS CRITERIA - I can...

1. Draw and predict before powering a circuit.
2. Place ammeters in series and voltmeters in parallel.
3. Compare measured and predicted values using evidence.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “ammeter” and “voltmeter” to explain your model or result. Sentence starters are optional.

ammeter: an instrument placed in series to measure current

voltmeter: an instrument placed across two points to measure voltage

trial: one run or repetition of a test

01 Bring back an earlier idea

An ideal series loop has supply 15 V. One component drops 3 V.
Find the other component’s drop.

Word help: series = connected along one unbranched current path

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: V = IR.
Use approved low voltage. Ammeter in series; voltmeter across a component. Disconnect before rewiring; never put an ammeter across the supply.

Worked reasoning: Supplied measurements show V=2, 4, 6 V and I=0.2, 0.4, 0.6 A. Each V/I is 10 ohms. This supports an ohmic model over the measured range.

Your turn: Plot V against I and mark the slope.
State why the data do not prove the model at every possible temperature or voltage.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

03 Look and explain

On a resistor circuit sketch, add an ammeter to measure loop current and a voltmeter to measure the resistor’s drop.

Sentence starter: The … in my model represents … . Its label shows … .



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Build the solution

A supplied 6 V, 12 Ω ideal circuit predicts I=6÷12. **Calculate** it. Then prepare a table for actual voltage, actual current and one observation; leave measurements blank until the trial.

Word help: ideal = using stated simplifying conditions

Sentence starter: I predict … . My model supports this because … .

Voltage (V)	Current (A)	Observation

Marzano check - circle: 1 2 3

05 Use the idea

A reading disagrees with your prediction. List two checks before replacing the model or deleting data.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: 1 2 3

06 Choose a method

An ideal blackbody’s kelvin temperature becomes 4 times as large. How does its peak wavelength change?

Word help: blackbody = an ideal absorber/emitter used in a thermal-radiation model

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

07 Exit check

Where must the teacher check your circuit before the first powered test?
Record that checkpoint and one change suggested.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A later reading is 8 V and 0.5 A. Does it fit the same 10 ohm model?
Name two checks before declaring the physical law wrong.

Sentence starter: The step I would change is … . The evidence is … .

Marzano check - circle: 1 2 3